

the following script and then access the data:

```
from sgmlib import SGMLParser
class selector(SGMLParser):
    def reset(self):
        SGMLParser.reset(self)
        self.data = []

    def handle_data(self, data):
        if data.count('.') >= 3:
            self.data.append(data.strip())

import urllib
def get_data(url):
    page = urllib.urlopen(url)
    parser = selector()
    parser.feed(page.read())
    parser.close()
    return parser.data

def get_ip():
    url = 'http://admin.password@192.168.1.1/hwhtml/summary/sum.html'
    return str(get_data(url))
```

The ideas behind the above code were discussed in last month's article on the subject. In the parser, we wrote a trivial method to extract any data that looks like a URL. The base URL is for a Huawei modem supplied by BSNL. The user name and the password can be passed as a part of the URL. The resulting output will look like what's shown below:

```
['D57.2.17', '59.94.245.130', '59.94.240.1', '255.0.0.0', '192.168.1.1',
'255.255.255.0']
```

In this case, the external IP address for the modem is 59.94.245.130. This is followed by the IP address of the gateway and the netmask.

By the way, in the last article, we had used *urllib2*. Authentication handling is a little more complex and flexible using *urllib2*; so, the above code used *urllib* instead and saved about five lines of code. In case you would like to explore authentication using *urllib2*, see www.voidspace.org.uk/python/articles/authentication.shtml.

Sending the IP address by e-mail

Since Gmail is very common, let us use that as the example to send the IP address to yourself. Here is the script you will need to write:

```
import smtplib
recip = 'youremail@yourdomain'
gmail_user = 'familymember@gmail.com'
gmail_pw = 'password'
smtpserver = smtplib.SMTP('smtp.gmail.com',587)
smtpserver.ehlo()
smtpserver.starttls()
smtpserver.ehlo()
smtpserver.login(gmail_user, gmail_pw)
hdr = "To: youremail@yourdomain
From: familymember@gmail.com
Subject: IP Address
"
msg = hdr + "\n" + get_ip() + "\n\n"
smtpserver.sendmail(gmail_user, recip, msg)
smtpserver.close()
```

You use the Python module *smtplib*. Gmail requires a secure connection and that the sender be authenticated. The complete message consists of a header with a blank line separating it from the message text. The text content is added by calling one of the *get_ip* methods you have written above.

You can use the *email* module to send mime attachments—for example, screen dumps—and create just the right tools to help you support your friends and relatives.

Ping the hosts

In case there is a network problem, you will need to narrow down to the cause. A home may have a netbook (or a laptop), a main desktop, a modem and wireless router. The netbook may be accessing media files on the main desktop. You can ping to each device

